

Project Proposal

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1. What problem am I solving? State it clearly.

Investigating causal drivers of neighborhood opposition (NIMBYism) to housing [Einstein et al., 2019, p. 28]. **Goal:** separate stable, **invariant** predictors of resistance from **spurious associations** driven by **environment-specific confounders** [Peters et al., 2018, p. 84]. **Aim:** find features predicting opposition consistently across the distinct **market regimes** of the 2018–2025 period.

2. Why is this problem important? To whom?

Critical for urban planners. Current models overfit to specific eras, failing when market regimes shift [Arjovsky et al., 2019, p. 1]. Identifying *invariant* causes enables policy design addressing fundamental concerns rather than symptoms of a moment, ensuring **robustness** [Bühlmann, 2020, p. 404].

3. What is my strategy for solving it?

Using **Invariant Risk Minimization (IRM)** [Arjovsky et al., 2019, p. 5] and **Bayesian Invariant Prediction (BIP)** [Wu et al., 2025b, p. 2]:

1. **Environments:** Partition data via **expanding window Cross-Validation** into **Training Environments** ($\mathcal{E}_{tr}$) and **Test Environments** ($\mathcal{E}_{te}$):

Window	Training ($\mathcal{E}_{tr}$)	Test ($\mathcal{E}_{te}$)
1	2018	2019–2025
2	2018–2019	2020–2025
...	...	...
N	2018–2024	2025

2. **Representation:** Extract features $\Phi(X)$ from property data and protest letters [Schölkopf et al., 2021, p. 15].
3. **Methods:** We implement the two core paradigms scheduled for **Weeks 6–7** of the seminar:
 - **Invariant Risk Minimization (IRM):** A *regularized optimization* paradigm to learn Φ such that a **fixed dummy classifier** $w = 1.0$ is simultaneously **optimal** across all $\mathcal{E}_{tr}$, **enforced** via a **gradient norm penalty** [Arjovsky et al., 2019, p. 5]:

$$\min_{\Phi} \sum_{e \in \mathcal{E}_{tr}} R^e(\Phi) + \lambda \|\nabla_w|_{w=1.0} R^e(w \cdot \Phi)\|^2$$

- **Bayesian Invariant Prediction (BIP):** A *probabilistic inference* paradigm defining a latent **feature selection vector** $z \in \{0, 1\}^p$; the **posterior** $p(z|D)$ **targets the true invariant features** [Wu et al., 2025b, p. 2].

4. Why is this solution good? Where does it fall short?

It falls short because it assumes an **invariant causal mechanism**. The invariance assumption fails if the underlying reasons for NIMBYism undergo **mechanism change** over time. For instance, neighborhood opposition may shift from traffic congestion fears to displacement concerns as demographics evolve.

5. How can I demonstrate that the solution works?

Train on expanding $\mathcal{E}_{tr}$, test on all **future horizons** (2019–2025) in $\mathcal{E}_{te}$. Successful Invariant Model demonstrates **predictive stability** through the **invariance of its AUC** score across all horizons, maintaining reliability where baselines fail under **distribution shift** [Krueger et al., 2021, p. 1].

References

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